

# Causal Inference & Difference-in-Differences

## Lecture 2

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## **Last Lecture: A Refresher**

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# Where we left off

Last time we discussed:

- Causality and the potential outcomes framework
- The fundamental problem of causal inference
- Selection bias

Let's revisit selection bias through a vivid recent example.

# COVID-19 vaccines: a puzzle from 2021

Early in the rollout, a striking pattern emerged in the data:

*Mortality rates among the vaccinated were **higher** than among the unvaccinated.*

Some commentators concluded the vaccine was harmful — or at best, useless.

Were they right?

# Who got vaccinated first?

The vaccine rollout was *not* random.

First in line:

- Elderly individuals
- Immunocompromised people
- Long-term care residents
- Frontline healthcare workers

These groups had **much higher baseline mortality risk** than the general population — with or without a vaccine.

# The naive comparison

|                     | Vaccinated | Unvaccinated |
|---------------------|------------|--------------|
| Avg. mortality rate | higher     | lower        |

Is this the causal effect of the vaccine?

# The naive comparison

|                     | Vaccinated | Unvaccinated |
|---------------------|------------|--------------|
| Avg. mortality rate | higher     | lower        |

Is this the causal effect of the vaccine?

**No.** The two groups differed dramatically *before* anyone got vaccinated.  
The comparison conflates:

- the effect of the vaccine
- the effect of being in a high-risk group to begin with

# COVID with stylized numbers

Imagine two age groups, with stylized monthly mortality rates:

|          | Vaccinated | Unvaccinated | Vax effect |
|----------|------------|--------------|------------|
| Under 65 | 0.05%      | 0.10%        | −0.05pp    |
| Over 65  | 1.00%      | 2.00%        | −1.00pp    |

*Within each age group, the vaccine halves mortality.*



## But aggregate the two groups...

Suppose 90% of the vaccinated are over-65, but only 10% of the unvaccinated are.

|                   | Vaccinated | Unvaccinated |
|-------------------|------------|--------------|
| % over-65         | 90%        | 10%          |
| Overall mortality | 0.91%      | 0.29%        |

The naive comparison shows vaccinated mortality is *three times higher*.

The vaccine cut mortality in half — but the data appears to say it tripled mortality. Selection bias entirely reverses the conclusion.

# Selection bias, formally

What we observe:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$$

Decomposes as:

$$\underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{\text{causal effect (ATT)}} + \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{selection bias}}$$

**Selection bias:** the treated and untreated groups would have differed even *without* treatment.

For COVID:  $E[Y_i(0) | \text{vaccinated}] \gg E[Y_i(0) | \text{unvaccinated}]$ .

## More everyday examples

Selection bias is everywhere:

- **Hospitals look deadly.** People who go to hospitals are sicker than people who don't.
- **Gym-goers are healthier.** But did the gym *cause* the health, or do healthier people gravitate to gyms?
- **Harvard graduates earn more.** Is that Harvard, or the kind of person who gets into Harvard?

The pattern: the people who get the “treatment” differ systematically from those who don't.

# The randomization fix

Random assignment makes treatment status independent of potential outcomes:

$$E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0]$$

⇒ selection bias vanishes.

This is why randomized controlled trials are the gold standard.

But what if we can't randomize?

## Pause: what is the counterfactual?

The **counterfactual** is what *would have happened* to the treated unit had it not been treated.

We never observe it directly. Every causal method is, at heart, a strategy for **constructing** a credible counterfactual.

| Method | What it uses as the counterfactual         |
|--------|--------------------------------------------|
| RCT    | The randomly-assigned control group        |
| DiD    | The control group's <i>trend</i>           |
| RD     | Units just on the other side of the cutoff |
| IV     | Compliers under no-treatment               |

**Causal inference = building counterfactuals.**

## **When You Can and Can't Randomize**

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## A/B testing: RCTs at industrial scale

Tech companies run *thousands* of RCTs per year:

- Netflix: which thumbnail makes you click?
- Amazon: does this checkout button color sell more?
- Meta: does this newsfeed algorithm increase engagement?
- Uber: does this driver incentive reduce wait times?

Users are randomly assigned to “A” or “B”. Differences in outcomes are causal effects.

This works because the firm *controls* the assignment.

# When randomization isn't possible

Many of the most important questions can't be answered with an RCT:

- Did raising the minimum wage reduce employment?
- Did the carbon tax reduce emissions?
- Did expanding Medicaid improve health?
- Did the new policing strategy reduce crime?

You can't randomly assign minimum wages to states. You can't randomize a tax.

We need a method that works when treatment isn't randomly assigned.



# Natural experiments

Sometimes policy changes affect *some* units and not others.

- One state raises its minimum wage; a neighbor doesn't.
- One country expands a vaccine program; another doesn't.
- One city implements congestion pricing; others don't.

This isn't an RCT — assignment isn't random — but it's a *natural experiment*.

Difference-in-differences is the workhorse tool for analyzing them.

## **Card & Krueger (1994): The Setup**

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# Effect of minimum wages on employment

Theory is ambiguous:

- **Competitive labor market:** a binding minimum wage *decreases* employment
- **Monopsonistic labor market:** a binding minimum wage *can increase* employment

Even if the direction were unambiguous, theory can't tell us the size of the effect.

**This is an empirical question.**

# The ideal experiment

What we'd want:

- Randomly assign labor markets to treatment (minimum wage raised) and control (kept constant)
- Compare outcomes

What we actually get from a policy change:

- Some areas raise the wage, others don't
- Treatment is *not* randomly assigned
- But the comparison is still useful — if we make the right assumptions

## Card & Krueger (1994)

On April 1, 1992:

- NJ raised its minimum wage from \$4.25 to \$5.05
- PA's minimum wage stayed at \$4.25

Card and Krueger surveyed 473 fast-food restaurants:

- Wave 1: February/March 1992 (before)
- Wave 2: November/December 1992 (after)

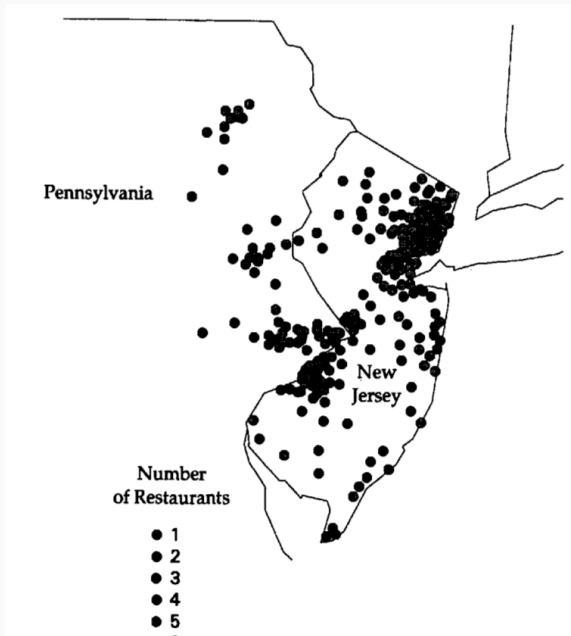
Outcome: full-time-equivalent (FTE) employment per restaurant.

# Why fast food? Why these states?

- Fast food employs many minimum-wage workers
- Standardized industry — restaurants are comparable
- NJ and eastern PA are **geographically and economically similar**
- Same labor market region, same business cycle, similar demographics

The geographic proximity makes PA a credible counterfactual for NJ.

## Restaurants in the sample



## The $2 \times 2$ Table

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## Average employment by state and period

|                  | PA              | NJ              |
|------------------|-----------------|-----------------|
| Before (Feb/Mar) | 23.33<br>(1.35) | 20.44<br>(0.51) |
| After (Nov/Dec)  | 21.17<br>(0.94) | 21.03<br>(0.52) |

Standard errors in parentheses.

## A first comparison

|        | PA    | NJ    |
|--------|-------|-------|
| Before | 23.33 | 20.44 |
| After  | 21.17 | 21.03 |

Is  $NJ_{\text{after}} - PA_{\text{after}} = 21.03 - 21.17 = -0.14$  a good estimate of the causal effect?

## A first comparison

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Is  $NJ_{\text{after}} - PA_{\text{after}} = 21.03 - 21.17 = -0.14$  a good estimate of the causal effect?

**No.** NJ and PA differed before treatment. This conflates the wage effect with pre-existing differences.

## A second comparison

|        | PA    | NJ    |
|--------|-------|-------|
| Before | 23.33 | 20.44 |
| After  | 21.17 | 21.03 |

Is  $NJ_{\text{after}} - NJ_{\text{before}} = 21.03 - 20.44 = 0.59$  a good estimate?

## A second comparison

|        | PA    | NJ    |
|--------|-------|-------|
| Before | 23.33 | 20.44 |
| After  | 21.17 | 21.03 |

Is  $NJ_{\text{after}} - NJ_{\text{before}} = 21.03 - 20.44 = 0.59$  a good estimate?

**No.** Many things change between Feb and Nov: weather, business cycle, holidays. We can't separate the wage effect from everything else.

# The DiD idea

Each naive comparison fails for its own reason:

- NJ vs PA *after*: groups differ in levels
- NJ before vs NJ after: time itself confounds the comparison

**Key insight:** compare *changes*, not levels.

- PA's change tells us what would have happened to NJ absent the policy
- Subtract that off — whatever's left is the policy effect

## The 2×2 table, completed

|        | PA    | NJ    | NJ – PA |
|--------|-------|-------|---------|
| Before | 23.33 | 20.44 | –2.89   |
| After  | 21.17 | 21.03 | –0.14   |
| Change | –2.16 | 0.59  | 2.76    |

$$\hat{\tau}_{DiD} = (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$$

PA employment *fell*; NJ employment *rose*. The DiD estimate is positive.

## Is 2.76 statistically significant?

The DiD estimate: 2.76 with standard error 1.36.

$$t = \frac{2.76}{1.36} \approx 2.03$$

A  $t$ -statistic of about 2 is the conventional threshold for statistical significance at the 5% level.

- Just barely “significant” — not a precise zero, not a sharply estimated positive effect
- 95% confidence interval roughly [0.1, 5.4]
- The data *rule out* a large negative effect

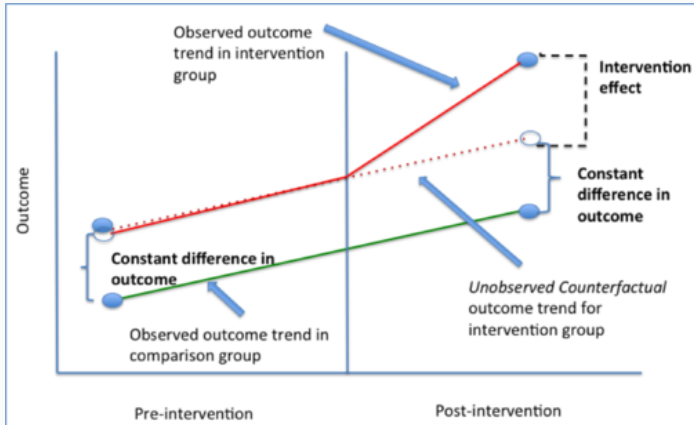
**Reading DiD tables:** the bottom-right cell is the estimate, parentheses are SEs, and the  $t$ -stat is the ratio.



# Card & Krueger Table 3

|                                                         |                 | <i>Restaurants,<br/>by State</i> |                 | <i>Difference</i> |
|---------------------------------------------------------|-----------------|----------------------------------|-----------------|-------------------|
|                                                         | <i>All</i>      | <i>PA</i>                        | <i>NJ</i>       | <i>NJ - PA</i>    |
|                                                         | (1)             | (2)                              | (3)             | (4)               |
| 1. FTE Employment Before,<br>All Available Observations | 21.00<br>(0.49) | 23.33<br>(1.35)                  | 20.44<br>(0.51) | -2.89<br>(1.44)   |
| 2. FTE Employment After, All<br>Available Observations  | 21.05<br>(0.46) | 21.17<br>(0.94)                  | 21.03<br>(0.52) | -0.14<br>(1.07)   |
| 3. Change in Mean FTE<br>Employment                     | 0.05<br>(0.50)  | -2.16<br>(1.25)                  | 0.59<br>(0.54)  | 2.76<br>(1.36)    |

# Visualizing DiD



Pre-treatment: a constant difference between groups. Post-treatment: DiD assumes that constant difference *would have continued* (the counterfactual). The intervention effect is the gap between the observed treated outcome and that counterfactual.

# Identifying Assumptions

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# What is an identifying assumption?

An **identifying assumption** is what you must believe for a method to recover the causal effect.

It's almost never directly testable — it's an argument you make.

Examples we'll meet this term:

- RCT: random assignment
- DiD: **parallel trends**
- RD: continuity at the cutoff
- IV: exclusion restriction

# The parallel trends assumption

**Parallel trends:** in the absence of treatment, treated and control would have followed the *same trend* over time.

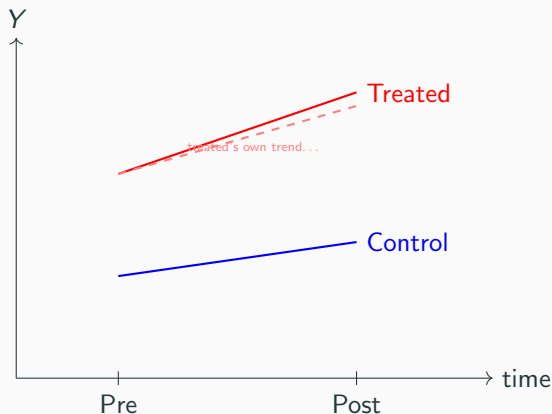
Formally:

$$E[Y_{it}(0) - Y_{i,t-1}(0) \mid \text{Treated}] = E[Y_{it}(0) - Y_{i,t-1}(0) \mid \text{Control}]$$

What we are *not* assuming:

- Levels can differ across groups (NJ and PA can have different baseline employment)
- Each group can have its own time-invariant features

# When parallel trends fails



If the treated group was already on a different trajectory, DiD attributes that pre-existing divergence to the treatment.

# Why might parallel trends hold here?

For Card & Krueger:

- NJ and eastern PA share the same regional economy
- Same business cycle, same weather, similar consumer base
- Fast-food industry is structurally similar across the border

The geographic proximity is the *argument* for parallel trends.

This is what credible DiD designs look like: a story for why control units would have moved like treated units, absent the policy.

# Assessing parallel trends

You can't test the assumption directly (it's about a counterfactual), but you can build credibility:

1. **Pre-period balance.** Were treated and control similar before treatment?
2. **Pre-trends.** Were they moving in parallel before treatment?
3. **Placebo tests.** Apply DiD to a period or group with no real treatment.



# Pre-period balance in C&K

## Means of Key Variables

|  |            | Restaurants, by State |                     | t-test<br>for<br>NJ - PA <sup>a</sup> |
|--|------------|-----------------------|---------------------|---------------------------------------|
|  | All<br>(1) | New Jersey<br>(2)     | Pennsylvania<br>(3) | (4)                                   |

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|                                         |      |      |      |      |
|-----------------------------------------|------|------|------|------|
| 1. Distribution of Restaurant Types (%) |      |      |      |      |
| a. Burger King                          | 41.7 | 41.1 | 44.3 | -0.5 |
| b. KFC                                  | 19.5 | 20.5 | 15.2 | 1.2  |
| c. Roy Rogers                           | 24.2 | 24.8 | 21.5 | 0.6  |
| d. Wendy's                              | 14.6 | 13.6 | 19.0 | -1.1 |
| e. Company Owned                        | 34.4 | 34.1 | 35.4 | -0.2 |

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|                                   |                |                |                |      |
|-----------------------------------|----------------|----------------|----------------|------|
| 2. Means in Wave 1                |                |                |                |      |
| a. FTE Employment                 | 21.0<br>(0.49) | 20.4<br>(0.51) | 23.3<br>(1.35) | -2.0 |
| b. Percent Full-Time<br>Employees | 33.3<br>(1.2)  | 32.8<br>(1.3)  | 35.0<br>(2.7)  | -0.7 |
| c. Starting Wage (\$/hr)          | 4.62<br>(0.02) | 4.61<br>(0.02) | 4.63<br>(0.04) | -0.4 |
| d. Wage = \$4.25 (%)              | 31.0<br>(2.3)  | 30.5<br>(2.5)  | 32.9<br>(5.3)  | -0.4 |
| e. Price of Full Meal<br>(\$)     | 3.29<br>(0.03) | 3.35<br>(0.04) | 3.04<br>(0.07) | 4.0  |
| f. Hours Open                     | 14.4<br>(2.4)  | 14.4<br>(2.5)  | 14.5<br>(2.5)  | -0.3 |

## Reading the balance table

NJ and PA restaurants in the pre-period look similar on:

- Chain composition (Burger King, KFC, Wendy's shares)
- Starting wage (\$4.61 vs \$4.63)
- Hours open per weekday
- Recruiting bonuses

NJ had *lower* FTE employment to begin with — but levels are allowed to differ. What matters is parallel *trends*, not parallel levels.

The similarity on observables is reassuring, though not a guarantee.

## **Did the Policy Bite?**

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## Wage distribution before the policy

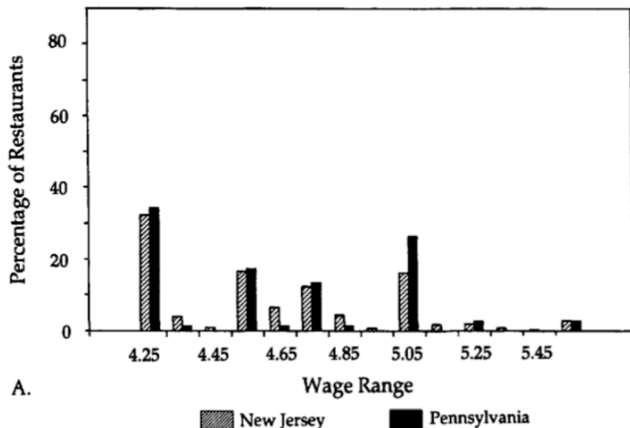


Figure 2.2 Distribution of starting wage rates. A. February–March 1992.  
B. November–December 1992.

Both states cluster around the \$4.25 federal minimum.

## Wage distribution after the policy

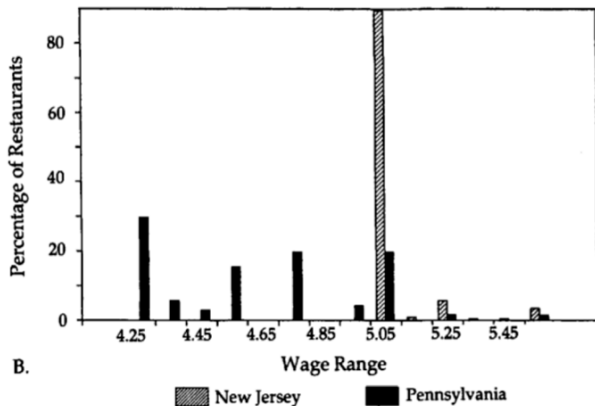


Figure 2.2 Distribution of starting wage rates. A. February–March 1992.  
B. November–December 1992.

NJ now has a massive spike at \$5.05 — the new minimum. The policy clearly bound.

# Why this matters

The wage distribution figures show:

- NJ restaurants actually *complied* with the new minimum
- There is meaningful “treatment” to detect
- PA’s distribution barely changed — a clean control

**First-stage check:** before asking whether the policy affected employment, confirm the policy actually changed wages. It did.

## DiD as OLS

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# DiD is just a regression

The DiD estimator can be recovered from OLS. One simple form, on *changes*:

$$\Delta y_i = \beta_0 + \beta_1 \text{NJ}_i + \varepsilon_i$$

where  $\Delta y_i$  is the change in employment at restaurant  $i$ .

$\hat{\beta}_1$  is the difference-in-differences.



# Why this works

For PA restaurants ( $NJ_i = 0$ ):

$$E[\Delta y_i \mid \text{PA}] = \beta_0$$

For NJ restaurants ( $NJ_i = 1$ ):

$$E[\Delta y_i \mid \text{NJ}] = \beta_0 + \beta_1$$

$$\Rightarrow \beta_1 = E[\Delta y_i \mid \text{NJ}] - E[\Delta y_i \mid \text{PA}] = \Delta_{NJ} - \Delta_{PA}$$

Difference of changes = difference-in-differences.

## Equivalent regression in levels

Equivalently, with restaurant-level data in both periods:

$$y_{it} = \alpha + \beta \text{NJ}_i + \gamma \text{Post}_t + \delta (\text{NJ}_i \times \text{Post}_t) + \varepsilon_{it}$$

|    | Pre              | Post                               |
|----|------------------|------------------------------------|
| PA | $\alpha$         | $\alpha + \gamma$                  |
| NJ | $\alpha + \beta$ | $\alpha + \beta + \gamma + \delta$ |

The interaction coefficient  $\delta$  is the DiD estimator.

# Connection to randomization

## Under random assignment:

- No selection bias to begin with
- A simple cross-sectional regression  $Y_i = \alpha + \beta D_i + \varepsilon_i$  identifies the ATE
- No pre-period needed

## Without randomization but with parallel trends:

- Selection bias is present in *levels*
- But it cancels out when we compare *changes*
- The interaction term  $\delta$  recovers the causal effect

Parallel trends buys back what randomization would have given us for free.

# Adding controls

The regression form lets us include controls:

$$\Delta y_i = \beta_0 + \beta_1 \text{NJ}_i + \mathbf{X}'_i \gamma + \varepsilon_i$$

Useful for:

- Holding restaurant characteristics fixed (chain, ownership, region)
- Reducing residual variance, tightening standard errors
- Checking whether the DiD estimate is sensitive to specification

## C&K regression results

|                                                     | Dependent Variable:<br>Change in Employment <sup>a</sup> |                |                 |                 |                 |
|-----------------------------------------------------|----------------------------------------------------------|----------------|-----------------|-----------------|-----------------|
|                                                     | (1)                                                      | (2)            | (3)             | (4)             | (5)             |
| 1. New Jersey Dummy                                 | 2.33<br>(1.19)                                           | 2.30<br>(1.20) | —               | —               | —               |
| 2. Initial Wage Gap <sup>d</sup>                    | —                                                        | —              | 15.65<br>(6.08) | 14.92<br>(6.21) | 11.98<br>(7.42) |
| 3. Controls for Chain<br>and Ownership <sup>e</sup> | No                                                       | Yes            | No              | Yes             | Yes             |
| 4. Controls for Region <sup>f</sup>                 | No                                                       | No             | No              | No              | Yes             |
| 5. Standard Error of<br>Regression                  | 8.79                                                     | 8.78           | 8.76            | 8.76            | 8.75            |
| 6. Probability Value for<br>Controls <sup>g</sup>   | —                                                        | 0.34           | —               | 0.44            | 0.40            |

## Reading the regression table

- Column 1: NJ dummy coefficient is 2.33, SE 1.19 — the basic DiD estimate
- Column 2: with chain/ownership controls, 2.30, SE 1.20 — barely changes
- Columns 3–5: alternative specifications using initial wage gap

Stability across specifications is itself reassuring. If your headline number jumps when you add a control, something's off.

The coefficient is positive and roughly 1.5 to 2 standard errors from zero. Not a precise zero, not a sharp negative effect.

# What did C&K find?

The minimum wage increase *did not* reduce employment. If anything, employment rose slightly in NJ relative to PA.

This was a **frontal challenge to textbook economics**. The competitive labor market model predicts a clear negative effect — and the data refused to deliver one.

The paper:

- Reshaped how economists think about low-wage labor markets
- Helped earn Card the 2021 Nobel Prize
- Launched a 30-year empirical literature

**Econometrics isn't just technique — it's how economists actually settle disagreements.**

## But C&K isn't infallible

The clean design has a flip side. NJ and PA are right next to each other.

### Spillovers:

- PA workers might cross into NJ chasing higher wages, reducing PA labor supply
- Customers near the border might cross into PA for cheaper food, hurting NJ demand
- Either would *contaminate* the control group

### Other concerns:

- Survey data on employment is noisy (later studies using payroll data found different results)
- One short window, one industry, two states — generalizability is limited
- Restaurants might adjust on margins other than headcount (hours, prices, training)



## **Beyond Two States, Two Periods**

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# What if we have more states? More years?

Card & Krueger is the cleanest possible DiD: 2 states, 2 periods, 1 policy change.

Real policy questions are often messier:

- Many states, many years
- Different states change policy at different times
- Policy changes vary in size
- Multiple outcomes, multiple subgroups

DiD generalizes — and we'll spend the next few lectures unpacking how. For now, two previews.

## Preview 1: Addison, Blackburn, & Cotti (2009)

**Question:** Did minimum wage changes affect retail employment?

**Setting:**

- US data, 1990–2005
- Many counties, many years
- Minimum wages varied across states and over time

**Key feature:** this isn't  $2 \times 2$  anymore. They use *panel data* with county and year fixed effects — a generalization we'll cover next lecture.

# Addison et al. results (illustrative)

**Table 4**

Regression estimates with county-level trends: food and beverage store sector.

| Industry                         | Food and<br>beverage<br>stores | Supermarkets<br>and other<br>grocery stores | Convenience<br>stores | Specialty<br>food<br>stores | Beer, wine,<br>and liquor<br>stores |
|----------------------------------|--------------------------------|---------------------------------------------|-----------------------|-----------------------------|-------------------------------------|
| NAICS code                       | 445                            | 44511                                       | 44512                 | 4452                        | 4453                                |
| <i>Dep. variable</i>             | <i>(b) Employment</i>          |                                             |                       |                             |                                     |
| Minimum wage                     | 0.225**<br>(0.114)             | 0.210*<br>(0.128)                           | 0.227**<br>(0.116)    | 0.067<br>(0.110)            | 0.101**<br>(0.048)                  |
| Population                       | 0.414**<br>(0.154)             | 0.526**<br>(0.231)                          | 0.821<br>(0.581)      | -0.693<br>(0.521)           | -0.035<br>(0.273)                   |
| Total employment                 | 0.468**<br>(0.053)             | 0.520**<br>(0.090)                          | 0.509**<br>(0.138)    | 0.903**<br>(0.176)          | 0.468**<br>(0.112)                  |
| Total average<br>weekly earnings | -0.052<br>(0.032)              | -0.043<br>(0.056)                           | -0.122**<br>(0.060)   | -0.179**<br>(0.058)         | -0.036<br>(0.033)                   |
| Unemployment rate                | -0.002<br>(0.003)              | -0.003<br>(0.004)                           | -0.006<br>(0.008)     | -0.005<br>(0.008)           | -0.000<br>(0.005)                   |
| Enrollment rate                  | -0.052<br>(0.032)              | -0.057<br>(0.056)                           | 0.050<br>(0.067)      | -0.219**<br>(0.064)         | -0.032<br>(0.038)                   |

# What to take from Addison et al.

- Estimates across food/beverage retail subsectors
- Coefficients suggest minimum wages did *not* reduce employment in this sector — some are positive
- Don't worry about the table's details now; the point is that DiD scales up

**Foreshadowing:** fixed effects and panel data are how we run DiD with many groups and many periods. That's next lecture.

## Preview 2: Clemens & Strain (2018)

**Question:** Did the minimum wage increases of 2013–2015 affect employment of low-skilled workers?

**Setting:**

- July 2009: federal minimum wage rose to \$7.25
- Then states diverged — some raised more, some indexed to inflation, some did nothing
- Average increase: \$0.92 (12%)

**Key feature:** multiple “policy regimes,” not just treated vs control.

**TABLE 1**

**List of States with Statutory Minimum Wage  
Increases and Inflation-Indexed Increases**

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**Statutory Increase of \$1  
or More**

Alaska  
California  
District of Columbia  
Massachusetts  
New Jersey  
New York  
Rhode Island  
South Dakota

**Statutory Increase  
Under \$1**

Arkansas  
Connecticut  
Delaware  
Hawaii  
Maryland  
Michigan  
Minnesota  
Nebraska  
West Virginia

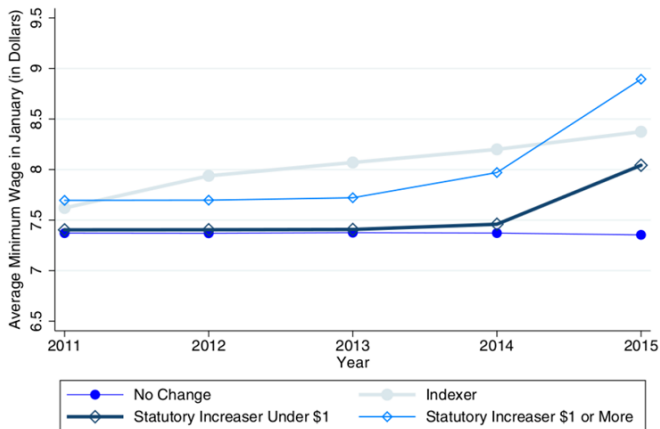
**Indexers**

Arizona  
Colorado  
Florida  
Missouri  
Montana  
Ohio  
Oregon  
Vermont  
Washington

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# Minimum wage trajectories

## Average Minimum Wage across Policy Regimes

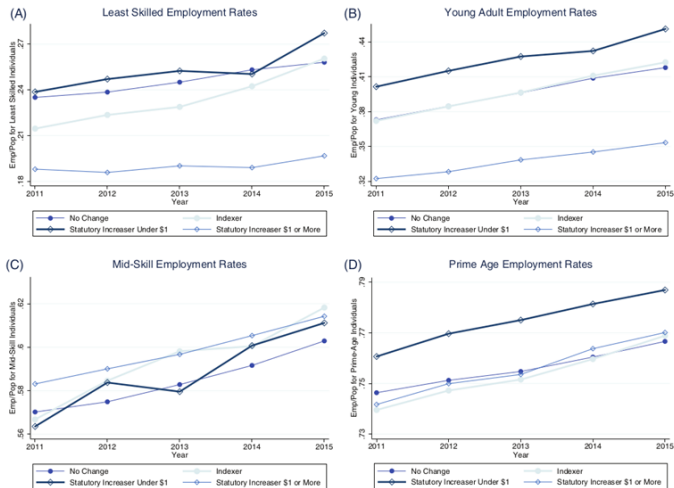


Different “treatment intensities” — not just on/off.



# Employment trajectories by regime

**FIGURE 3**  
Evolution of Employment across Minimum Wage Policy Regimes



Notice: employment trajectories visibly differ across regimes for low-skilled

# The Clemens & Strain regression

$$y_{ist} = \sum_p \beta_p \text{Policy}_s \times \text{Post}_t + \alpha_{1s} + \alpha_{2t} + X'_{ist} \gamma + \varepsilon_{ist}$$

## Subscripts:

- $i$  = individual,  $s$  = state,  $t$  = year

## Each piece:

- $y_{ist}$ : outcome (employment status of individual  $i$  in state  $s$  in year  $t$ )
- $\text{Policy}_s \times \text{Post}_t$ : state's policy regime  $\times$  post-change indicator
- $\sum_p \beta_p$ : separate DiD coefficient for each policy regime  $p$  (large increaser, small increaser, indexer)
- $\alpha_{1s}$ : **state fixed effect** — absorbs anything time-invariant about a state
- $\alpha_{2t}$ : **year fixed effect** — absorbs national shocks common to all states
- $X'_{ist} \gamma$ : individual-level controls

## From C&K to C&S

The C&S regression *is* DiD — just generalized:

- C&K had **one** treated group and **one** post period
- C&S has **many** states (some treated, some not, in different ways) and **many** years
- State fixed effects ( $\alpha_{1s}$ ) play the role of the “Treated” dummy
- Year fixed effects ( $\alpha_{2t}$ ) play the role of the “Post” dummy
- The interaction  $\text{Policy}_s \times \text{Post}_t$  is still the DiD coefficient

**Same idea, scaled up.** We'll unpack fixed effects properly next lecture.

**TABLE 4**  
Relationship between Minimum Wage Increases and Employment of Individuals 25 and under with  
Less than a High School Education (D-in-D Estimates)

|                                                | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
|------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Large statutory increaser $\times$ post        | -0.0106<br>(0.008)  | -0.0137<br>(0.008)  | -0.0104<br>(0.011)  | -0.0105<br>(0.007)  | -0.0098<br>(0.007)  | -0.0118<br>(0.009)  |
| Small statutory increaser $\times$ post        | 0.0113<br>(0.013)   | 0.0123<br>(0.012)   | 0.0113<br>(0.013)   | 0.0108<br>(0.012)   | 0.0091<br>(0.012)   | 0.0094<br>(0.010)   |
| Indexer $\times$ post                          | 0.0203**<br>(0.008) | 0.0199**<br>(0.009) | 0.0204**<br>(0.009) | 0.0180**<br>(0.008) | 0.0161**<br>(0.008) | 0.0137*<br>(0.008)  |
| Ln(income per capita)                          |                     | 0.1477<br>(0.101)   |                     |                     |                     | 0.1223<br>(0.099)   |
| Housing price index divided by 1,000           |                     |                     | -0.0044<br>(0.086)  |                     |                     | -0.0072<br>(0.087)  |
| State mid-skill employment-to-population ratio |                     |                     |                     | 0.2521**<br>(0.125) |                     | 0.2443**<br>(0.121) |
| Age and education controls                     | No                  | No                  | No                  | No                  | Yes                 | Yes                 |
| Observations                                   | 453,956             | 453,956             | 453,956             | 453,956             | 453,956             | 453,956             |
| R <sup>2</sup>                                 | .015                | .015                | .015                | .015                | .101                | .102                |

*Notes:* This table reports difference-in-differences estimates for which the policy indicator variables distinguish between states in which the minimum wage was increased by less than \$1 and states that increased their minimum wage by \$1 or more. The sample is from the ACS and includes individuals ages 25 and younger with less than a completed high school education. Variable definitions and sources are discussed in the note to Table 2 (and in the paper). All specifications include year and state fixed effects. Age and education controls consist of a dummy variable for each education group and age (included in columns 5 and 6 as indicated within the table). Standard errors are clustered at the state level.

\* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

# What to take from Clemens & Strain

- Different policy regimes have different effects
- “Large statutory increasers” show negative point estimates for low-skilled employment
- “Indexers” show positive estimates
- DiD with multiple treatment intensities — a richer design than C&K

The minimum wage debate is ongoing. Different designs, different samples, different periods can yield different conclusions. That's the empirical literature, not a failure of method.

## Threats and Recap

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# Threats to DiD

Even with parallel trends, DiD can mislead if:

- **Anticipation:** treated units change behavior *before* the policy
- **Spillovers:** the control group is indirectly affected (PA workers cross to NJ)
- **Compositional changes:** who is in each group shifts over time
- **Differential trends:** treated and control were already diverging

Each is a story you have to argue isn't happening.

## Aside: another Card classic — the Mariel Boatlift

**Question:** Does immigration lower native wages?

**Setting (Card 1990):**

- April–October 1980: 125,000 Cubans arrived in Miami in a sudden wave
- Miami's labor force grew by  $\sim 7\%$  almost overnight
- Comparable US cities (Atlanta, Houston, Los Angeles, Tampa) saw no such shock

**Design:** DiD, with Miami as treated and similar cities as control.

**Finding:** no detectable negative effect on native wages or employment, even for low-skilled workers.

Same Card, same method, totally different setting. DiD is a workhorse, not a one-off.



# Where DiD is used

DiD is one of the most widely used empirical methods in:

- **Labor economics:** minimum wage, immigration, unemployment insurance
- **Health economics:** Medicaid expansions, smoking bans, drug approvals
- **Education:** school reforms, charter schools, tuition policy
- **Environmental:** cap-and-trade, emissions standards, carbon taxes
- **Industry:** geo experiments, marketing attribution, feature rollouts
- **Government:** ex-post evaluation of tax changes, aid programs, regulations

Anywhere a policy or change rolls out unevenly across units and time.

# Recap

1. Selection bias is the central problem of observational causal inference — and the COVID example shows how easy it is to be fooled.
2. RCTs (and A/B tests) solve it by randomizing. Policy questions usually can't be randomized.
3. DiD compares *changes* across treated and control groups.
4. Identifying assumption: **parallel trends**.
5. DiD is just OLS with a treated dummy, post dummy, and their interaction.
6. Parallel trends does the work that randomization does in an RCT.

**Panel data and fixed effects:** generalizing DiD to many groups and many periods.

We'll see how fixed effects naturally extend the C&K logic, and how studies like Addison et al. and Clemens & Strain fit into that framework.

Reading: Angrist & Pischke (2014) Chapter 5; Stock & Watson Ch. 13.4.

# References

- Addison, J. T., Blackburn, M. L., & Cotti, C. D. (2009). Do minimum wages raise employment? Evidence from the U.S. retail-trade sector. *Labour Economics*, 16(4), 397–408.
- Card, D., & Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. *The American Economic Review*, 84(4), 772–793.
- Clemens, J., & Strain, M. R. (2018). The short-run employment effects of recent minimum wage changes: Evidence from the American Community Survey. *Contemporary Economic Policy*.

## Further reading:

- Angrist, J. D., & Pischke, J.-S. (2014). *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press. (Chapter 5)
- Card, D. (1990). The impact of the Mariel Boatlift on the Miami labor market. *ILR Review*, 43(2), 245–257.